

Board Games as a Research Method

A Case Study on Research Game Design and Use in Studying Algorithmic Mediation

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ABSTRACT

Investigation of board games as a research phenomenon has a long tradition in the academic community. However, using them as a research method has not been explored as much. This case study presents the steps towards the development of a serious board game aimed at collecting data on how participants understand algorithmic mediation in academic social media platforms. We followed the Serious Game Design Assessment (SGDA) framework to build a game through several playtests. Challenges in this process are presented in terms of the research game's Content, Mechanics, Fiction, and Aesthetics. We discuss how we addressed these challenges and elaborate on the different types of data that we collected by means of our research game. Finally, we present the lessons learned, highlighting the ideal target group for playtesting, the need of simplicity, the importance of storytelling, and how to counter the risk of participants' strategies to "game" the game.

CCS CONCEPTS

• **Participatory design**; • **HCI design and evaluation methods**;
• **Collaborative Interaction**;

KEYWORDS

Board games, serious games, SGDA framework, Algorithmic mediation

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1 INTRODUCTION

Board games are traditionally studied in academic literature as a research phenomenon. See, for instance, studies on the resurgence of board games to disconnect from the internet [7], or studies on historical board games to understand human history and psychology [1, 10]. However, the potential of board games as a research method has not been explored as much, and there is a lack of literature about the types of data that board games can generate and the main challenges in designing them [9, 23]. We chose to design and use board games as a creative research method to collect insights on the ways in which users of social platforms make sense of algorithmic mediation in an academic context. Algorithmic mediation is the selection, prioritisation, and display of personalised content in digital platforms performed by algorithms. These processes of choosing and showing personalised content to users require a combination of automated decision-making techniques, such as user profiling, content indexation, the identification of patterns in browsing history and inference of relevance [17, 19]. Platforms such as ResearchGate and Academia.edu, which we call Academic Social Media (ASM), use algorithmic mediation to recommend different types of content, papers to read, researchers to follow and jobs to apply for. These platforms are socio-technical artefacts that shape and are shaped by people's perceptions of their features [6, 12, 18]. Given the incipient literature on the ways in which users of such platforms make sense of algorithmic mediation in their work, leaving many topics still unexplored, we set out to inquire both our process – yielding lessons learned from crafting the board game as a research method – as well as our outcomes – in terms of data aimed at generating a better understanding of how people make sense of algorithmic mediation in ASM platforms. In this case study, part of the research project Algorithmic Mediation in Academic Social Systems (AMASS)¹, we (a) introduce the concept of a research game; (b) present how we selected and put together the elements of the game; (c) discuss the design process from the first prototype until the final version of the game; (d) show different kinds of output this method generates and how these data can be analysed; and (e) present our learnings

¹To learn more about the project, kindly visit: <https://soc.kuleuven.be/mintlab/blog/project/amass/>.

that can help other researchers to use custom made board games as a research method.

2 RESEARCH GAMES FOR ELICITING USERS' PERCEPTIONS OF NEW TECHNOLOGIES

Choosing a physical board game as a method to study technology, algorithmic mediation and digital platforms might be unconventional. However, board games present several characteristics that make them very suitable for understanding people's perceptions of new technologies and their applications. First of all, games usually contain narratives [14], which allow participants to envision futuristic elements (such as new technologies) without questioning how feasible those elements are. This way, "games might create fictional, but natural settings for players to explore, use, and discuss such futuristic elements" [9]. Second, board game mechanics typically ensure that all participants can have their say. Given that each player takes their turn to participate, an organised flow of engagement is ensured, which differs from the typical dynamics of focus groups, for instance [23]. Third, games constitute an icebreaker, leaving participants at ease and creating a pleasant experience for them. Because games are likely to create a playful and relaxed atmosphere, groups of people who do not know each other tend to interact more easily [9]. Fourth, while playing games, participants are less likely to worry about what the researcher expects them to answer. The focus will usually turn to the goal of winning the game, therefore the data collected is more faithful to what the participants really think rather than what they think the researcher wishes to hear.

3 INITIAL DESIGN CHOICES

We chose the elements of our board game based on two main goals. Firstly, we wanted to find out how collaboration and competitiveness are reflected in peoples' choices when using certain features of ASM platforms. Secondly, we aimed to understand how researchers relate human values to algorithmic recommendation features in these platforms.

Due to the inherent competitiveness in the academic environment, researchers strategically consider how to make their research output public and accessible, for instance complying with the principles of Open Access before the research is published [16]. Although early-career researchers hold great appreciation for a sharing culture, "they are concerned about sharing unpublished papers, interim results, data, and ideas because of competition and the possibility of people stealing information/ideas and making them their own" [20]. It seems that the need for recognition that enables funding, academic positions and status within the scientific community leads to specific communicative strategies that are often shaped by the ways scientific capital [3] is measured. Therefore, we see many intertwined and sometimes contradictory factors, such as competition and collaboration, sharing and hiding information, motivations driving researcher's decisions, and algorithmic mediation selecting and prioritizing content in the user's feed. To understand the impact of these factors on users' perceptions of academic social media, we incorporated all of them in the board game.

Next to accounting for competition and collaboration, we also included human values, as people's decisions can have a personal

or a social focus and are typically motivated by their values. Human values (e.g., benevolence or power) are "the vocabulary used to express the goals in social interaction" [21]. We relied on the categorization of values theoretically and empirically validated in 20 countries with great cultural variety [21].

We used the Serious Game Design Assessment (SGDA) framework [14] to inform the design of the research game, as it offers a structured approach for designing the various elements of games that have a 'serious' purpose, such as collecting research data. Previous work has shown that this framework also serves to design games on top of assessing them. Geerts and colleagues used SGDA framework to inform the design of research games in three different domains, namely train information systems, hospital logistics and future TV watching scenarios (Geerts et al., 2018).

The SGDA framework consists of six elements: purpose, content, framing, mechanics, fiction, and aesthetics. The **purpose** of a serious game is always twofold because it combines both the *aim* of the players within the game, as well as the *impact* expected by the game designer, which in the case of a research game is the intended research goal. The **content** means the information that the game contains, which can be featured in the cards, in the board, the narrative, etc. **Framing** addresses play literacy, the game's topic, and its target group. **Mechanics** refer to the learning curve, rules, and goals that the players will follow. **Fiction** involves the narrative (e.g., plot and characters) within the game. And finally, **aesthetics** refers to the graphics of the game setting, style, colours, and visualisation features.

To provide a realistic context, we designed a research game that emulates recurrent situations in scholarly communication, where researchers interact with each other and are faced with challenges they need to overcome using ASM. The overall gameplay revolves around academic tasks (e.g., writing a paper) for which players are invited to use cards to select how they would complete them. From their cards, each player chooses an approach – which can be "collaborative", "competitive" or "ambivalent" –, one or two recommendation strategies and one or two motivations for their decisions (both recommendation strategies and motivations are explained in section 4.1). Then, players are instructed to present their choices to the other players, seeking to receive votes (in the form of citations) from the other players and providing arguments for their own decisions. After that, participants anonymously vote for the presentation they like better, also justifying their vote. After four rounds, the player with the most citations (i.e., votes received) wins the game.

4 DESIGNING THE AMASS BOARD GAME USING THE SGDA FRAMEWORK

The board game is the result of an iterative design process that involved careful planning, development, prototyping and playtesting. During this process, we performed five face-to-face playtests with the physical board game and one online playtest using a digital version, involving participants that are part of the target group. Following the SGDA framework helped us to adjust, after each round of playtesting, each element separately while keeping what worked in the prototype. Some of these elements needed more than one adjustment, while others remained the same throughout the

entire development. In the following paragraphs, we present the most relevant choices and changes made during the game's design process concerning the game's Content & Information, Mechanics, Fiction & Narrative, and Aesthetics & Graphics.

4.1 Content & Information

This component of the game refers to “information, facts and data visible to the player” [14]. For instance, in Monopoly the game content consists of properties, railroads, utilities, a jail, and various other instances of interaction between the players, such as the money used to pay the rent, the “towns” represented on the board, etc. In the AMASS game, the content is mainly presented by means of cards, which are directly linked to our research goals. We designed three decks of cards: **motivation cards**, based on psychology literature about human values; **strategy cards**, resulting from a platform analysis we conducted earlier [17, 19]; and **task cards**, inspired by literature on scholarly communication. Motivation cards and task cards were redesigned as a result of playtests.

4.1.1 Motivation cards. To understand which values our participants associate with algorithmic recommendations, we designed a total of 10 cards, each containing a human value identified by Schwartz [21]: Achievement, Benevolence, Conformity, Hedonism, Power, Security, Self-Direction, Stimulation, Tradition, and Universalism. In the playtests, two issues appeared. First, players seemed to have different interpretations of the values, as the concepts were not properly explained. To solve this, we added definitions to the values in the cards, as they are described in the literature (see Figure 1 (b)).

Second, our playtests showed that people were hesitant to call these cards “value cards” because they saw values as strong and permanent. Theorists from psychology and sociology view values as “the criteria people use to select and justify actions and to evaluate people (including the self) and events” [22]. Values motivate people's behaviour and are situated, meaning they acquire relative importance according to each specific situation. Thus, we changed the name of the cards to “motivation cards” and explained that motivations can change depending on the situation. Participants were then asked to rank the cards in order of importance for each round. This change made it easier for participants to dissociate the motivations for a specific action from their own personal values.

4.1.2 Strategy cards. During the game, players are instructed to associate human values with algorithmic features, using the strategy cards. These cards represent the behaviour of the digital platform, i.e., the content curation carried out by algorithms based on users' online actions. Cards design is based on an analysis of ASM carried out in previous stages of the research², and resulted in 9 cards with features such as “matching topics”, “people I follow” and “publications I've read” (see Figure 1 (c)).

4.1.3 Task cards. To associate algorithmic features with human values, players should picture themselves conducting certain academic activities, represented by 4 task cards (see Figure 1 (a)). Each

task card contains different activities related to scholarly communication, such as “write a paper” and “select a journal to publish your work”. For the content of these cards, we drew inspiration from scientific practices described in the “Model of Scientific Communication of a Global Distributed Information System” [2]. We started by mapping every task that can be performed through ASM platforms.

The first version of the task cards listed simple actions, such as “Preserve your publication” or “Analyse and draw a conclusion”, phrased as faithful as possible to the theoretical model. However, during the playtests, players reported it was hard to grasp the full meaning of the tasks, as they lacked context and were not connected to larger activities as in the original model. To improve the understandability, and encourage roleplay, we introduced storytelling into the cards by also adding contextual factors from the model. These contextual factors included previous or future scientific tasks and the role that the researcher fulfils during different parts of the process [2], namely “reader”, “writer”, “disseminator” or “evaluator”. Adding this allowed the players to picture themselves in the situation of performing the task. During the following playtests, we noticed more consistency in the interpretation of tasks by participants. In total, we designed four task cards because, according to our playtests, more than four tasks would make the activity too tiring for the participants.

4.2 Mechanics

Game mechanics define the possible actions for players within a game, which are expressed through the game rules. For example, in Monopoly, game mechanics consist mainly of throwing a dice to move forward and earning money from players who have to pay the rent when ending up at your property. Besides the pivotal in-game aim of the players, the SGDA Framework also highlights the importance of “the operation of the reward system, the main playful obstacles/challenges, and the difficulty balancing and the win condition” [14]. The first version of our board game was based on the “Game of Life”, with a timeline starting from graduation, publications, citations, events, societal impact, reaching academic positions, acquiring more funding for projects, receiving recognition from the academic community and finally on a trajectory towards the Nobel Prize. This version presented two main issues that were solved with a change in the game mechanics.

Firstly, in this first version of the game, people needed to guess, secretly, what actions the other players would choose, and the players would reveal their guesses all at the same time. This moment of revelation caused great commotion during the playtesting: people would celebrate when they guessed right or argue when they didn't. This created a problem for the serious purpose of the game (collecting research data), because we wanted to hear every single voice while people were reasoning to justify their cards. To avoid overlap in audio recordings, we changed the game mechanics, so players no longer have to guess each other's strategy, but rather defend their own strategy to their peers. In turns, players would then present their chosen approach and strategies, substantiating their choices with motivation cards and explaining the reasoning behind it. After all the presentations, the players vote anonymously for which strategy they think is the best to solve the task at hand.

²The platform analysis in this project aimed at describing the algorithmic mediation in ASM. Four analyses were carried out: an interface analysis, a web page code inspection, a patent analysis, and a company inquiry. More about this process can be found in [17, 19].

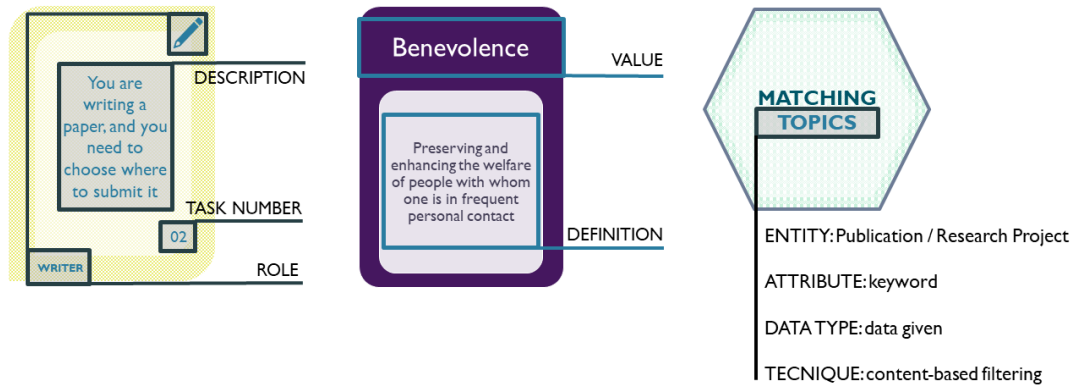


Figure 1: – (a) Task cards explained; (b) Motivation cards explained; (c) Strategy cards explained

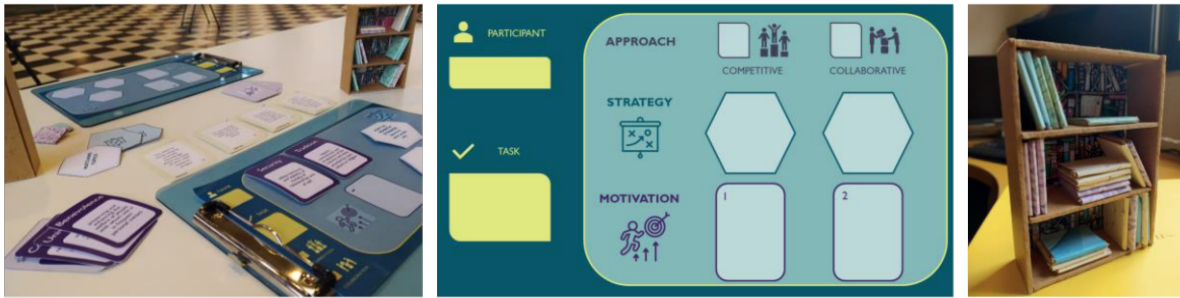


Figure 2: – (a) Set-up of the physical game; (b) Player's individual board (plan of action); (c) Bookshelf and books (representing votes)

The players cannot vote for themselves and must justify their votes. The votes are sent via written private message to the moderator, who counts the votes. At the end of the round, the players receive a miniature book (see Figure 2 (c)) representing each vote they received. The game ends once all task cards are played. The participant who collects the largest number of books (representing votes) at the end of the game wins it.

Secondly, the initial reward system was problematic because it had a shared board where everyone could see the other players' tokens. What we learned in the playtests is that because the points were based on votes, this format could lead to people refraining from voting for someone who was winning, regardless of the merit of their strategy. The initial reward system thus allowed players to vote for someone else just to prevent that person from winning. To counter this problem, we created individual boards (see Figure 2 (b)) and a hidden reward system in the form of a bookshelf (see Figure 2 (c)), which meant votes would be revealed only at the end. This new mechanic ensured a more organised dynamic where all the participants had their turn to present their ideas and justify their votes individually in secret to the moderator. The data collected from these interactions was also easier to analyse than the data from the earlier playtests.

4.3 Fiction & Narrative

The fictional context of a game refers to the “setting, narrative, story, scenario, characters, back story, problem, and so on for the game play” [5]. For instance, in Monopoly you are a landowner that aims to gain as much land as possible. In the final version of our board game, participants are researchers who impersonate different roles performed by researchers in real life whether by writing and disseminating one's own publications or reading and evaluating others' work. Players are challenged (through task cards), to come up with viable solutions (using strategy cards) that are motivated by human values (on motivation cards), and get rewarded (with citations) by convincing their peers of the value of their ideas. This dynamic was designed to resemble some real-life academic interactions. The role on the task cards represents the different “hats” researchers typically wear during their academic path and are to be considered by the participants while solving the tasks. To solve the task, participants present their strategy (using strategy cards) and the motivation (motivation cards) behind their choices. This presentation emulates scientific conferences, where researchers present their research results. Normally in these venues there is an opportunity for interactions, providing and receiving feedback to/from peers, and this moment is represented in the game by the voting phase, during which participants cast a vote for the best solution to the task, and explain why they believe that solution is

Table 1: - Participants

ID	Gender	Experience	Field	Position	Age	Nationality
SABE_P1	F	4-6 years	Social Sciences	PostDoc researcher	18-30	Australia
SABE_P2	M	4-6 years	Social Sciences	PostDoc researcher	31-40	The Netherlands
SABE_P3	F	1-3 years	Social Sciences	PhD Student	18-30	Belgium
SABE_P4	Non-binary	1-3 years	Social Sciences	PhD Student	18-30	Italy
SBBE_P1	F	7-9 years	Natural Sciences	PostDoc researcher	31-40	Spain
SBBE_P2	M	1-3 years	Engineering	PhD Student	18-30	Spain
SBBE_P3	M	1-3 years	Engineering	PhD Student	18-30	Spain
SABR_P1	F	4-6 years	Humanities	PostDoc researcher	31-40	Brazil
SABR_P2	F	4-6 years	Social Sciences	PhD Student	31-40	Brazil
SABR_P3	M	4-6 years	Humanities	PhD Student	31-40	Brazil
SBBR_P1	F	4-6 years	Natural Sciences	PostDoc researcher	31-40	Brazil
SBBR_P2	F	1-3 years	Medicine	PhD Student	41-50	Brazil
SBBR_P3	M	4-6 years	Natural Sciences	PhD Student	18-30	Brazil

the best. Peer recognition is a defining characteristic of scientific communities [11]. Recognizing the quality and merit of researchers' work is commonly done through citations, which are considered a form of scientific reward [13]. In our research game, we mirrored the scientific reward system by giving each participant a bookshelf, to which the player can add citations in the form of miniaturised physical books each time someone votes for their plan of action (see Figure 2 (c)). Citations are also widely used as indicators of scientific activity, as they constitute the calculus in university rankings, journal quality assessments and researchers' reputation, including in ASM [15].

4.4 Aesthetics & Graphics

All the elements of the game (content, fiction, target group, setting and mechanics) are framed by formal aspects reflected in the aesthetic of the game [14]. The choice of colours, fonts, images, visual identity, and overall appearance of the game elements can communicate a particular atmosphere, intended by game designers. For example, the Mr. Monopoly character in the Monopoly game evokes the idea of a successful businessman. In this sense, aesthetics can evoke desirable emotional responses in the player, when interacting with the game. In the first prototype, the board and the cards had vibrant primary colours, simple and plain shapes, with no depth, which -as we reasoned afterwards- could be considered slightly childish. Therefore, as a serious game is meant to be played by adult researchers, we decided to use a more elegant colour pallet and change the shapes of the cards. In the final version of the board game, we used cold tones (purple, blue-green, grey) and matte shades to reflect professionalism in line with the target group (framing) and the fact that the game resembles daily research practices (i.e. its fiction and narrative). This made the overall aesthetics more polished (see the game's set-up in Figure 2 (a)). Whilst keeping the "serious" colour palette, we appealed to playfulness with cute little books representing the citations. We hoped that this way, these reward elements could create a joyful and calm experience and alleviate the potentially stressful decision-making embedded in the game dynamics. The books and bookshelves are

handmade using recycled material, which is less austere and dry than a score board (See Figure 2 (c)).

5 ANALYSING DATA FROM THE AMASS BOARD GAME

We performed 4 game sessions with a total of 13 participants, as shown in Table 1. After the data collection, we analysed the data in two different processes, one process for each research question. We briefly present some of the quantitative and qualitative data we collected during the board game sessions rather than presenting the full results, as this paper focuses on how we designed the board game and used it for data collection.

5.1 Quantitative data

We used the cards or positions on the board as variables, quantifying the participant's moves. By counting the actions, we gained insights into the participants' preferred choices. For instance, we could easily see the most picked card or the most chosen alternative to draw conclusions from this data. In the game, participants in most situations had chosen the collaborative approach.

Using the game elements as variables also makes it possible to combine different elements. In the game, we asked players to choose between different cards in a deck and associate these cards with other cards to show how they would perform a certain task. Counting how frequently these cards are put together can show a tendency of association. For example, we quantified the number of times someone chose a particular combination of cards, such as the strategy card "matching authors" and the motivation card "power". With the associations mapped, we drew a data visualisation in the form of a network. In Figure 3, the line suggests the strength of the association: the more people chose that particular association between two cards, the thicker the line connecting the cards. The reasons why these associations are strong, however, should be investigated in more depth, preferably with qualitative methods, as we show in subsection 5.2.

Another possible analysis strategy is picking a specific niche to understand less popular preferences. This analysis can substantiate the development of personas or archetypal profiles that will inform

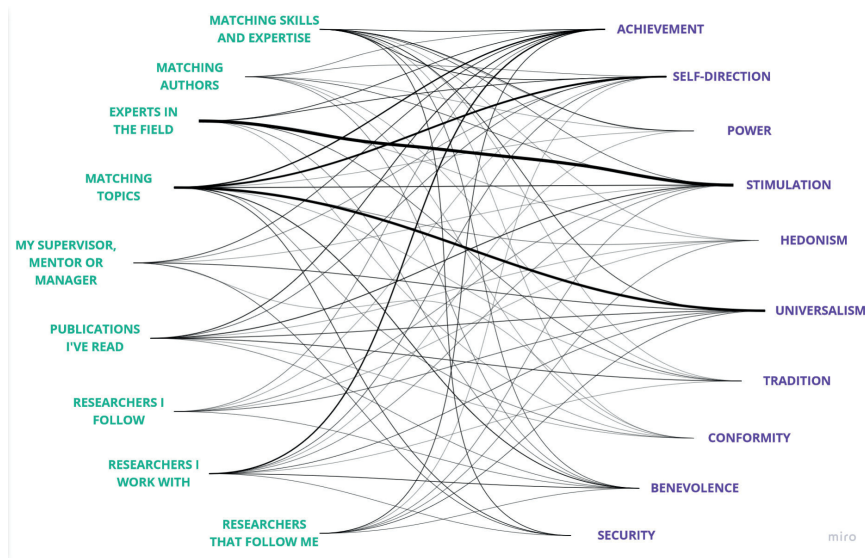


Figure 3: - Human values versus Recommendation Strategies

design decisions. For example, the game players often chose the collaborative approach to solve the tasks, which could overshadow the other alternatives (namely, competitive, and ambivalent approaches). By isolating the less popular choices, it was possible to see a certain profile in more detail and visualise how the choice of approach impacts the rest of that participant's choices, such as the values, recommendation strategies, and the group's demographic characteristics. This analysis showed that some characteristics are notably different depending on the chosen approach, which was useful to know which characteristics were relevant to integrate the archetypal profile. The analysis of all the profiling characteristics based on cards choices resulted in four archetypal profiles, called: the collaborative reader, the competitive writer, the collaborative disseminator, and the ambivalent evaluator. The four evidence-based archetypal profiles represent people's choices in the game. Friedman and colleagues [8] affirm that systems designers necessarily convey social and moral values in their work. The archetypal profiles offer support for future research and for value-sensitive platform design [8], as they show the values and recommendation strategies people mobilise when executing specific tasks.

5.2 Qualitative data

Games can also generate several types of data to be examined from a qualitative perspective. In our case, we began by separating the transcriptions of the participants' verbalisations during the gameplay per session, participant and per round of the game, so we had the excerpt of what participant X said during their presentation of task Y on session Z. Then, we labelled these excerpts (in the NVivo software) with the card combinations the excerpts were referring to. This description allowed us to gather all the excerpts that referred to a certain combination of cards. For instance, we were able to group all the excerpts where participants explained why they chose the cards "matching authors" and "power" in the same round.

Next, we performed a thematic analysis [4]. After reading the excerpts, we created inductive open codes representing the participants' reasons for choosing the card combinations, such as "challenge the *status quo*" and "keywords help scan literature". We then went back and forth through the sets of excerpts to find patterns among the participants' reasonings. We concluded the thematic analysis by organising the codes around meaningful themes, such as "fair and egalitarian science" and "desire to overcome the experts" and included quotes from the participants in the qualitative analysis. Through this analysis, we could understand in depth the connections participants made between human values, algorithmic features and their approaches. For example, the association between 'stimulation' and 'experts in the field' shows two tendencies: One is admiration and respect for the specialists, and the other is the desire to overcome the experts, not mirroring them, but rather distinguishing themselves from those perceived as authorities in the field. One of the participants said: *'Usually, I have a little more of a competitive look at the experts in the field. So, I look at the experts and see what they're doing, how they're doing it. And then I start to look at what they are not doing'*. Participants feel stimulated by the challenge of working with something new and finding problems to work on that other researchers did not solve yet.

6 LESSONS LEARNED

The key lessons learned from our case study on the design and use of a research game as a method to engage participants in sharing their reasoning about recommendations in ASM platforms, are formulated in six actionable recommendations that we believe are broadly applicable to research games used in a variety of research inquiries:

- *The quality of board game should be assessed from both a research and a game perspective:* While research games have some similarities to focus groups or co-design workshops, it is a game, which means one needs to playtest it in an

iterative way and not only run a pilot test (as we typically do to test and improve a research protocol and its research instruments). Besides, the alignment between the game and the research should also be assessed.

- *Adjust the language to your players:* Avoid jargon or language that your players are not familiar with. The appropriateness of the language can be assessed during the playtesting with people from the same target group of the research project.
- *Adjust the game mechanics in light of your data collection:* The main goal of a research game is to collect research data, and to not invest too much time explaining the game rules or risk invalidating data due to misinterpretation of the rules. It is therefore crucial to design game rules that are easy to understand after a simple and short explanation. To do so, you can (a) create a short video with instructions and ask participants to watch it in advance; (b) play a start round for trying out the gameplay and to give participants the chance to try it before committing to the real game; and (c) make sure to answer all questions from participants before the game starts.
- *Contextualise the game:* Create a narrative that combines the different aspects of your game into a coherent, meaningful whole. Consider adding (a) role(s) to be “played” to encourage players to empathise and fully explore the narrative and properties of the game. Using humour or aesthetics can make the game engaging.
- *Anticipate on how players might game the system:* thinking about the meta game and all possible ways a player can game the system (in this case, the research game) is extremely important to get valid data. An interesting idea is to invite an experienced board game player to try to hack it, to find vulnerabilities in the game elements.

7 CONCLUSION

The case study presented in this paper shows the potential of using tangible board games in highly technological projects, where the object of study are algorithms on digital platforms. The case study gives insight ‘behind the scenes’ of the iterative design process, including a transparent account of the main challenges and how they were dealt with.

We also show how board games have great potential to facilitate both quantitative and qualitative data gathering, as researchers can include elements they wish to investigate into the game. For instance, the moves of the players, the associations between different cards and the socio-demographic characteristics can work as variables for quantitative analysis. As for qualitative data, by including a moment for players to express their thoughts on the decisions they are making, researchers can go into the sensemaking of the participants and the reasons why people do what they do.

We encourage researchers to use board games considering the lessons we learned from the process. Future studies could include new ways to analyse data from research board games, or creating a new framework for research games ideation. By sharing their case studies, researchers can contribute to high-quality research on game creation processes and outcomes.

8 LIMITATIONS

We highlight that, although the results of this case study are specific for the context of algorithmic mediation, the lessons learned can apply to a variety of different domains. As exemplified in section 3, the SGDA framework was already applied to other research games, with fruitful results. We believe the outputs of this case study are building blocks that can bring helpful insights to others when designing games for research.

We also acknowledge that the game mechanics, which includes a rewarding system based on votes from other players and the mediation of the researcher herself, can lead to socially desirable outcomes. After all, it is still a group activity and the presence of the researcher can impact the participants’ answers towards socially desirable ones. On the other hand, the designed mechanic was intended to resemble the academic environment, and peer recognition is a pivotal part of the inner workings of academia.

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